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Contexts that promote interest, self-determination, and learning: lasting impressions and lingering questions

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Abstract

Across eight narratives, each capturing unique characteristics of a particular after-school or out-of-school computer program, we identified several unifying themes. We offer those themes as lingering questions that merit exploration by those invested in these ‘club’ settings. Specifically, we ask about the conceptions of learning held by authors, examine the distinctions between learning *about* and learning *with* technology, and question the degree of choice required to sustain the interest and engagement portrayed in these narratives. © 2000 Elsevier Science Ltd. All rights reserved.

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1. Introduction

Children dissect frogs, inventory and repair computers, design websites, create oral histories, publish original stories, play games, and chat on-line. Such are the variety of activities described in the eight articles that comprise this Special Issue of *Computers in Human Behavior* dedicated to technology-based after-school and out-of-school programs. Indeed, the stories that unfold in these pages offer varied glimpses of adult and student interactions around technology, interactions that take place away from the more regulated and constrained environments of classrooms and schools. We could not help but be caught up in the promises and challenges of technology in the rather untethered settings that contributing authors describe. The

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level of engagement and the degree of enthusiasm they often depict are infectious. These are among the lingering impressions that these authors left with us.

However, we must admit to being a bit like the parents and teachers that “sat on the sidelines of the room” in Gillingham and Youniss’ Computer Club. Technology still holds great mystery to us as educators, despite our daily encounters with websites, e-mails, data analysis, and word processing. As with those parents and teachers in the Computer Club, we remain somewhat on the technological periphery hoping to discover ways to use hypermedia to the betterment of our students, not just outside the classroom, but also inside.

Thankfully in this volume, we have the guidance of experts who have shown us what promises and pitfalls we might expect when students and technology come together and when students are permitted the luxury of self-directed exploration. We see the promises in these students’ displays of confidence and self-assurance, their enthusiasm, their questioning and persistence, and in their thoughtful reflections about themselves and others. We see the pitfalls in the adults’ uncertainty of their place in this configuration, and in their struggles to maintain the best of formal learning while endowing it with the freedom of informality. Above all, these experts have made it apparent that programs are not created in the mind or on paper, no matter what we believe about grant proposals, program handbooks, curriculum documents, or research designs (Bruce & Bruce; Garner, Zhao & Gillingham; Zhao, Mishra & Girod). Programs have a life of their own that only comes into being when students, adults, and environmental affordances move into alignment (Garner & Zhao). That is why the KLICK or Fifth Dimension programs transplanted to alternative states or different sites developed their own unique character.

Despite the uniqueness of these programs from California, Illinois, and Michigan, there are several themes that link them together. Two of these themes pertain directly to the concept of learning as depicted by the authors, and one relates to learner motivation in the guise of self-determination and interest. Our final theme deals with our continuing conundrum, the role of the educator in informal, after-school and out-of-school programs. We present these themes in the form of lingering questions we take away from these narratives.

2. Conceptualizations of learning

Every contributor to this Special Issue speaks about these out-of-the-classroom experiences in terms of student and adult learning. Whether learning was the figure or ground at these sites, it would appear that learning was nonetheless an intended outcome of these technology-based projects. Yet, while the word ‘learning’ is a familiar one to us, we found it difficult at times to uncover what the authors meant by learning. Similarly, claims that learning was occurring at program sites were not often accompanied by any particular explanation as to how such learning was substantiated. We are not suggesting that program developers and site supervisors must be of one mind when it comes to their views of learning, only that they put words to

those unspecified notions. For us, learning entails a discernible growth in students' knowledge, a deepening of their interests, and enhanced strategic abilities (Alexander, 1997; Alexander, Jetton & Kulikowich, 1995; Wade, Trathen & Schraw, 1990). The vagueness we noted in these articles may be indicative of the fact that traditional concepts of learning cannot be or should not be easily mapped onto these alternative settings. It may also be that a goal of learning has only a secondary or tertiary role to play in these after-school and out-of-school programs. But how do you document learning in these informal settings without interfering with the goals of the clubs?

To be fair, there were times when we could readily identify aspects of student learning in the authors' descriptions. These included the changes in participants' notions about science and scientists (Bruce & Bruce), their new-found ability to disassemble and reassemble hardware (Gillingham & Youniss), their computer-based problem solving (Grosshandler & Grosshandler), and their creation of multimedia products including video yearbooks, oral histories, newsletters, and web applications (Garner & Zhao; Garner, Tan & Zhao; Zhao et al.). Gallego and Cole described children's progress in English and Spanish literacy, and their understanding of academic concepts. Yet, at other times we found our conceptions of learning stretched almost beyond recognition. For instance, when Wong, Packard, Girod and Pugh cast participants' computer game playing or their creation of virtual selves in chat rooms as positive instances of learning, we became somewhat incredulous. Can we track the positive developments in participants' knowledge, interest, and problem solving in this situation?

We acknowledge that our conceptions of learning may simply be too limited to accommodate these more innovative cases, but it still begs the question as to what these experts perceived as learning in these informal contexts and how such learning was or should be documented. Are we to celebrate children's efforts to create virtual selves — not to be confused, in our judgment, with Markus and Nurius' (1986) concept of multiple selves? Should we see children's engrossment with video games on par with Csikszentmihalyi's (1990) construct of the optimal flow experience? Wong et al.'s reflection as to what degree optimal experience (ala Csikszentmihalyi) becomes optimal *educational* experience captures our puzzlement in this matter. Perhaps it is the case that any engagement and any outcome should be embraced as learning in these informal settings. Yet, we resonate with Gallego and Cole's observation that there is a great need to arrange for children, as part of these playful after-school programs, to engage in the kind of educational activity that benefits their current and future academic lives. Perhaps, the students themselves should define what learning is in these informal settings (Wong et al.). But, would these young participants judge the fruits of all these activities to be learning? That remains to be seen.

We wholeheartedly agree with Garner, Zhao and Gillingham's reminder that "What's happening is what matters" in these articles. For that reason, we would benefit from a clearer sense of what the authors, as project developers and scaffolders, conceive learning to be and whether this conception can survive the journey from informal to formal learning environments.

3. Learning about and learning with

In trying to make sense of the person–technology interplay, Salomon, Perkins and Globerson (1991) distinguish between the effects *of* and *with* technology. Although both have a role to play in learning and development, Salomon et al. see the effects *of* as the residues left behind when the engagement is complete. Effects *with*, by comparison, are the empowerment that the technology affords the individual during engagement. Salomon et al.'s (1991) contrast between effects *of* and effects *with* fits nicely with the stories told in this Special Issue. The promises we spoke of earlier are more evident when the authors are conveying this 'person-plus-technology' empowerment that Salomon et al. (1991) envision than when they are speaking in terms of the residue of that encounter.

As a way to frame students' pursuits of understanding in these after-school and out-of-school programs, we want to build on Salomon et al.'s (1991) *of* and *with* notions by comparing learning *about* and learning *with* technology. We would assume from these narratives that computer technology in these programs is seen principally as tools for students' learning and development. This is certainly one of the explicit goals of the KLINK projects (Zhao et al.). Thus, for those in four rural Michigan sites, technology holds the promise of new jobs and new opportunities (Garner & Zhao). Technology, in this case, becomes as Dewey (1913) says *means* as well as *ends*. In several of these programs, the goals of learning *with* are rather well articulated. Thus, in Project SEARCH, technology is a means to dismantle limited notions about science and scientists, and expand students' understanding of and attitude toward scientific exploration. For the Odyssey program, technology is a tool for critical thinking and problem-solving activities. Garner, Tan and Zhao describe technology in the service of writing a newsletter for the KLINK network. All are cases of learning *with*.

What is most interesting, however, is that these after-school and out-of-school programs seemed to work more effectively and the student accomplishments seemed easier to articulate when energies were focused on learning *about* rather than learning *with* technology. When the students finally lifted the cover off the computer and began to get inside the machinery in the Computer Club, or when they became involved in building websites or learning about spreadsheets in other programs, the power of these informal learning environments came into clearer focus. This perception may have resulted from several factors. For one, it may be the case that learning *about* technology is an initial step for some of these participants who have had precious little exposure to hypermedia. It is also conceivable that learning *about* is a more suitable activity for student-led computer clubs than learning *with*, since it builds on students' fascination with such technologies. Of course, it is possible that learning *with* brings with it some unwanted traces of traditional in-class learning that some of these participants would rather avoid (Bruce & Bruce). This is the frustration we felt in reading the Garner, Tan and Zhao article:

Given the ways in which the children responded — failing to complete stories, writing formal and fairly controlled prose, not bothering much with descriptive

detail or mechanics — we'd have to say that they seem to have written, when they wrote at all, for the same reasons that they write in school: an adult asked them to. (p. 46)

4. Motivational dimensions: self-determination and interest

Even more prevalent than conversations about learning, the authors of these articles paint a dramatic picture of student engagement and investment rarely witnessed during school hours (e.g. Gillingham & Youniss; Grosshandler & Grosshandler; Wong et al.). The authors attribute this engagement to such motivational factors as participants' self-determination, freedom of choice, opportunities for collaboration, and interest. We cannot quibble with this overall assessment. In many ways, it is difficult to perceive how programs based on willful participation can be anything but motivating when compared to traditional classrooms where attendance is mandatory, where tasks are assigned, and where their completion is demanded. However, we are still left with several lingering questions regarding the motivational character of such after-school and out-of-school programs.

For instance, self-determination, as we interpret the work of Deci (1992; Deci & Ryan, 1985, 1987) and others (Corno & Rohrkemper, 1985) represents individuals' belief that they have some say so or control of their lives. Opportunity for choice is considered one contributor to students' self-determination, but there are other contributors as well, including a downplay of external reinforcers and the presentation of rules and instruction as informational rather than as controlling (Ormond, 1998). Most importantly, self-determination is not conceived as an either/or condition but exists in varying degrees. Thus, even though it may be easier to spawn stronger feelings of self-determination in students participating in after-school and out-of-school programs, self-determination can be experienced even in the context of traditional classroom activities. In addition, self-determination does not require complete freedom of choice or the lack of any external controls.

Still, in the pages of these articles we sensed some trepidation at instituting any constraints or limitations on students' activities or desiring any specific educational outcomes:

KLICK! kids can learn as much or as little as they choose of whatever they choose. Some kids choose to only play games and they are not discouraged to do so. Choice is critical; in fact, evaluation studies suggest that without it, a KLICK! clubhouse struggles. (Wong et al., p. 314)

Consequently, we wondered how much freedom of choice contributing authors judged as necessary to maintain students' high level of engagement in these non-mandatory programs? Is it not enough that students are participants by their own volition? How much freedom or flexibility in program activities must be involved to sustain student self-determination or their attendance? Is it possible to give students too much freedom of choice in terms of program activities? Does the level of choice

pertain at all to students' foundational knowledge or personal interest, or do program directors already assume a rich base of knowledge about and personal interest in technology for all community members? It would appear that several of these programs are attempting to bring technology to student populations that have historically had limited access (Gallego & Cole; Gillingham & Youniss). Moreover, it would appear that program developers hope to sow the seeds of personal interest among students, rather than merely nurture deep-seated interests that already exist.

Our own research in interest suggests that those with personal interest in a domain will often value new knowledge and persist at demanding, abstruse, or complex tasks when those with less interest choose to walk away (Alexander & Murphy, 1998; Wade, Buxton & Kelly, 1999). In effect, the critical distinction between fun and learning that several authors describe may be potentially less relevant when one is interacting with those who are already invested in the domain at hand.

That several authors are so sensitive to maintaining a 'fun' atmosphere in the after-school programs raises certain issues. Is the character of these after-school and out-of-school programs such that it exacerbates the association between fun and learning? It may also be the case that some of the participants in these programs have only a 'fragile' interest in learning *about* or learning *with* technology. This fragile interest can be easily quashed when less situationally interesting tasks are encountered or when activities smack of schooled tasks. Such fragile interest might be even more of an issue when these after-school and out-of-school programs are intent on attracting more girls and minorities into their community (Grosshandler & Grosshandler).

As we weigh these motivational factors, it might benefit us to reflect on the following questions: To what degree do these programs succeed because they bring together participants who already feel self-determined in these more informal non-mandatory environments and who are already interested in learning *about* (if not *with*) technology? Conversely, to what degree do these programs contribute to feelings of self-determination and interest in their participants and how has this been documented? Finally, are the positive developments evidenced in these motivational factors particular to these informal settings or can they carry over into more formal learning environments?

5. The role of the educator in informal, after-school programs

In traditional school settings, teachers are positioned as authority figures in charge of regulating and evaluating students' activities and behavior. Even in alternative pedagogies that emphasize cognitive apprenticeship and scaffolded instruction, the focus has been on how the adult, as *the more knowledgeable other*, can facilitate children's progress in learning. In contrast to adult roles in school settings, many of the contributors to this volume describe situations in which children and adolescents either have more expertise than the adults or they wish to engage in activities without adult guidance. The result in many of these cases is uncertainty and role conflict for all the participants.

For example, students in the Computer Club, described in Gillingham and Youniss' article, developed technological expertise, which they offered as a service to the school. Although two of the teachers and the two consultants had the technological expertise to play traditional roles — such as providing technical support, structure to the initial meetings, help in problem solving, and encouragement — teachers and parents who did not possess technological expertise sat on the sidelines, “not sure whether their role was to teach or listen. They couldn't teach because they did not have the expertise as they did in their other subjects” (p. 261). In the Computer Club, it was the students who had become the more knowledgeable others.

Interestingly, attendance by the adults consistently dropped off. Given the difficulties that adults experienced in this study when they had less technological knowledge than students, we wonder how successful were similar projects such as the “Adopt-A-Teacher” program and the police tutoring programs of the KLINK! Clubhouses described in Zhao et al.

In other studies in this volume, adults who had either technological expertise or domain knowledge also experienced conflict, as they held onto traditional teaching roles and attempted to bring the culture of schools into the computer clubs. What follows is Zhao et al.'s description of some of the KLINK site coordinators, who adopted a didactic teaching approach:

They wanted to be the authority of knowledge, the person who knows everything. Thus they would not allow students to do anything that they did not know how to do themselves. This greatly limited the variety of activities students engaged in. Some site coordinators wanted to run KLINK! like a class. (p. 297)

Similarly, most of the university students described in Bruce and Bruce's article adopted the role of ‘lesson givers’, which was consistent with their views of teaching and schooling. As science teachers, the university students wanted children to learn content, not ‘play’, which they viewed as ‘nonproductive frivolity’. Zhao et al. contend that attempts to instill the culture of the school into the clubhouses interfered with student interest, self-determination, and learning. Likewise, Garner, Tan and Zhao found that an activity determined by an adult — specifically, writing like reporters for publication on the KLINK website — was largely unsuccessful because of the high failure-to-complete rate and the minimal, formal, and controlled writing that characterized most of the students' reports.

Conflicts are documented even in the after-school and out-of-school program where the adult considered self-determination to be essential to the program, sought flow experiences for students, and consciously attempted to stay out of the activities as much as possible (Grosshandler & Grosshandler). In this case, the adult clearly played a supportive, facilitating role and was sometimes a player himself. Yet, he too had rules. For example, students had to build five models from booklets. Like many of the other contributors to the volume and the educators they describe, Grosshandler and Grosshandler felt frustration when participants did not pay attention and were concerned about missing teachable moments, maintaining order,

allocating materials and attention, and supporting and recruiting more girls and students from under-represented groups.

Such uncertainty and conflicts in authority have been documented in studies of other kinds of voluntary after-school clubs such as after-school Read and Talk Clubs that took place in a public library setting (Alvermann, Young, Green & Wisenbaker, 1999). These discussion groups are analogous to the computer clubs in several ways: (1) they were voluntary; (2) students joined and stayed with the clubs because they were social outlets (although the students also received a small monetary reward); and (3) they were conducted in some ways that were different than school. Like the computer clubs, these discussion clubs were the site of role conflict. The adults felt it necessary to intervene when they were positioned as teacher or 'responsible adult' by the students. In such instances, they were expected to prevent long silences by asking questions, to keep the discussions on track, and to maintain order. Sometimes the adults felt it necessary to intervene on behalf of students who were excluded by other group members. Sometimes parents viewed the adults as role models. And, sometimes the adults wished the discussions to replicate the structure they were familiar with (i.e. discussing a common text), whereas some students wanted to read texts of their choice and discuss common themes. In discussions where no common text was discussed, the adult facilitators expressed reservations about the quality of the discussions.

The uncertainties and role conflicts these articles describe will become even more pronounced as the boundaries dissolve between school and after-school computer activities. Perhaps critiques of didactic teaching and the newer pedagogies such as instructional scaffolding and cognitive apprenticeships may help in resolving such dilemmas. Current views describe scaffolding as a complex, subtle, and dynamic interaction, and emphasize the importance of interpersonal relationships, which are affected by learners' developing personal and social identities and their desire, or resistance, to the goals of instruction (Goodnow, 1990; Rogoff, 1986; Stone, 1993). Critics argue that motivation in the zone of proximal development is not a child's desire for mastery itself but rather the desire to be like or to impress the adult. Thus, we need to re-examine not only what we are asking the learner to do but also "whom we are asking the learner to become" (Litowitz, 1993, p. 191). Since many of the adult participants in this volume view their role as facilitators or scaffolders of learning (at the minimum), future research might explore why some adults over-assist or resist turning over responsibility to the learner. How can teachers know how much assistance to give, especially to student groups who are diverse in culture and language? What conceptions of learning can we all agree on for after-school and out-of-school programs? And, how can play and learning be balanced or integrated in new conceptualizations of what it means to 'do school'.

6. Conclusion

This Special Issue of *Computers in Human Behavior* has described a variety of informal, voluntary, after-school and out-of-school computer clubs. Each of these

programs, as we have come to appreciate, is unique in its ‘localness’. The “starter kits” (Cole, 1996) that program developers provided were shaped by the surrounding communities, the school cultures, and by the goals and interests of participating students and adults. Still, we could identify threads of similarity woven throughout these compelling narratives. For instance, despite the metaphorical reference to clubs, all program developers retain an expressed learning agenda. As such, they struggle to differing degrees with the educational mission that came with their starter kits. Against this backdrop, we read the stories of club projects judged by the authors as successful along with those programs considered problematic. Implicit, and sometimes explicit, in such judgments are certain criteria defining learning, self-determination, and interest — the themes of this issue.

Adults who have assumed responsibility for these clubs have evaluated them in terms of students’ sustained participation and their learning, usually represented by the products of learning (e.g. informational websites, models, and tutoring programs). Children’s and adolescents’ input into these evaluations is most often through observations of their engagement, apathy, or resistance. In other words, participants voted with their feet by abandoning clubs that did not engage them, much like the adults on the sidelines in Gillingham and Youniss’ article. Or, they voted with their fingers on the keyboards, writing prodigiously in the chatrooms but failing to complete articles for the website or doing so with the least effort possible (Garner, Tan & Zhao).

What made the difference for participating children and adolescents? From our analysis of the articles in this Special Issue, we have concluded that students seemed to become the most engaged when they had choice, technological support when needed, and opportunities for meaningful discussion and interaction. These characteristics are consistent with Brown and Campione’s (1990) communities of learners model and Vygotsky’s (1978) notion of the zone of proximal development, in which social interaction and play both serve important functions. After-school programs — whether they involve technology or book discussions — *may* help move play into schooling. Or, more likely, play and schooling will continue to be dichotomized as it has always been. To quote Nigel Hall (1991): “Play was viewed by society at large as a noneducational process: it was all right for very young children and all right outside of school. School, however, was for learning” (p. 3). Even more disturbing and likely, given adults’ emphasis on learning outcomes and their traditional views of teaching, after-school and out-of-school programs may become more like school and less like play.

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